ChatFolio

Evaluation Report

Qualitative & Quantitative User Evaluation

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Qualitative Evaluation

Part 1

Study Overview & Participant Demographics

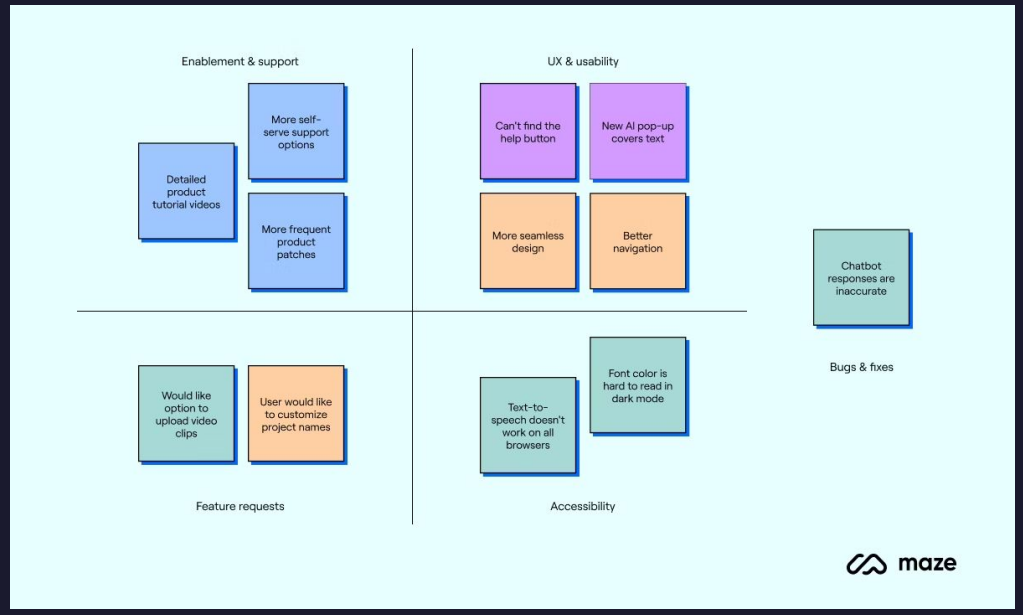
METHODOLOGY

Task: complete full ChatFolio flow
greeting → 4 questions → generate portfolio

Method: think-aloud, moderated observation

- Observer noted friction points, surprises, and positive reactions without guiding users

Synthesis: affinity diagramming → 6 themes



Study Overview & Participant Demographics… continued

PARTICIPANTS (n = 7)

- College students
- Majors: CS, Finance, Economics, or Liberal Arts

- Mixed investing experience:
  - Some had zero prior knowledge
  - Some were familiar with stocks/markets
  - Some were already actively investing

Affinity Diagram Results

Onboarding & Initial Prompt 6 / 7 groups

- Initial prompt is too prescriptive - suggests "house" or "retirement" as the only examples, boxing users in.
- Users felt like they had to fit into a predefined category even if their goal was more general (e.g. "just grow my money").
- Bot should ask about age, income, debt, and life stage *before* jumping to goal selection.
- "Additional Info" section at the end of the flow feels like an afterthought - that context should be gathered upfront.
- Better opener: "Tell me about yourself and what you're hoping to do with your money" instead of listing specific goals.
- Bot jumped straight to timeline questions before understanding the user's life situation.
- Users had to volunteer age, job, and income info themselves - the bot never proactively asked.
- When multiple goals were typed, model only addressed the first one.
- Example goals (retirement, house) skew older - not relatable for 22-year-olds just starting to invest.
- "It should feel like a conversation with a financial advisor who wants the whole picture."
- Need: prompts for projected income, rent, existing savings, student debt, and upcoming life changes.

Risk Tolerance Confusion 3 / 7 groups

- "35% confident" is a confusing and unusual metric - users didn't know how to interpret it.
- Risk level descriptions aren't sufficient for beginner investors - they need more context.
- Needs a visual spectrum or labeled scale (e.g. "Conservative → Aggressive") explaining each level.
- Newcomers fundamentally don't know what "risk" means in an investing context.
- Should ask "How experienced are you with stocks or finances?" early in the flow to calibrate language and recommendations.

Chat & Conversation Flow 5 / 7 groups

- Users dumped all their inputs into the first message, bypassing the intended step-by-step flow entirely.
- Chat persists after portfolio creation - users were unsure if it was for follow-up questions or to restart from scratch.
- Follow-up responses after portfolio generation were abrupt and provided insufficient feedback.
- When a user asked the chatbot to explain the "additional info" field, it skipped the explanation and jumped straight to portfolio generation.
- It's not clear to users that the chatbot can assist them during the profile generation process.
- Follow-up questions from the bot used financial jargon that newcomers didn't understand.
- No visual indication that sending a follow-up message triggered a portfolio regeneration - output appears with no transition.
- "More feedback if you send another chat - it's too abrupt and not clear it produced a new output."

Affinity Diagram Results… continued

Portfolio Output & Comprehension (5 / 7 groups)

- ETF tickers (VTI, VXUS, etc.) are confusing for beginner investors - most don't know what they are.
- Users don't know what "indexes" mean or where the specific funds come from.
- Portfolio percentage breakdowns are confusing - hard to connect to real money or to their specific goal.
- Users want a deeper dive into what each index fund does and why it's in their portfolio.
- Only Vanguard funds shown - users felt the selection wasn't diverse enough.
- Should allow users to set a target dollar amount (e.g. $1M) instead of just a vague physical goal.
- Hard to understand how the portfolio connects to a goal as specific as buying a house in a particular city.
- ETFs and financial terminology should be defined inline - not hidden in dropdowns or tooltips.

UX & Visual Polish (4 / 7 groups)

- Page auto-scrolls to the bottom during portfolio generation - users have to scroll back up to see the beginning.
- Text color when referencing user input values was inconsistent with the rest of the chat UI.
- "How to Get Started" section is hidden in a dropdown - it should be visible by default for new users.
- No loading indicator when a follow-up message triggers portfolio regeneration - users don't know if anything is happening.
- Mandatory vs. optional input fields are not clearly differentiated - users unsure what to fill in "Additional Info."
- Post-generation content is unclear: users don't know if additional follow-up questions are changing the portfolio or not.

What Worked Well (6 / 7 groups)

- Loading screen was helpful - users appreciated knowing the system was processing and not frozen.
- Sidebar updating in real-time as users answered questions was widely appreciated.
- Chart is clear, optimistic, and intuitive - users could read it without explanation.
- Portfolio explanation section was well-received once it loaded - users liked seeing the reasoning.
- "What is ChatFolio?" disclaimer was useful - users who read it upfront felt more oriented.
- Conversational format overall was liked - users preferred it to a traditional form.
- Users appreciated that the model visibly reflected their inputs in its recommendation.
- Data presentation is clean and easy to follow - diversification mix made sense once explained.

Key Findings by Theme

PAIN POINTS

→ **Onboarding**: Initial prompt implied only "retirement" or "house"; users felt their goals were being put in a box

→ **Risk Tolerance**: Beginners don't understand what "risk" means in investing; "35% confident" phrasing was confusing

→ **Chat Flow**: Users dumped all info in one message, bypassing the conversational step-by-step design

→ **Portfolio Output**: ETF tickers (VTI, VXUS, BND) and "index fund" were meaningless to beginners - literacy gap

→ **UX**: No visual indicator when a follow-up message triggers portfolio regeneration; no loading state shown

WHAT WORKED

✓ Real-time sidebar profile updates

✓ Loading screen / onboarding panel

✓ Growth projection chart (clear & intuitive)

✓ Conversational format strongly preferred over traditional forms

UX Takeaways

1 Gather life context before investment goals

Collect age, income, debt, and life stage before asking about specific goals — users need the system to understand them first

2 Close the financial literacy gap

Define ETFs, indexes, and risk tolerance inline. Never assume baseline financial knowledge for a beginner-focused tool

3 Fix the follow-up feedback loop

Users need a clear signal that their follow-up message was received and that the portfolio is regenerating

4 Lean into the conversational format

The conversational design is the core differentiator - invest in making it feel like a real advisor, not a form in disguise

Quantitative Evaluation

Part 2

The Prompt

What were the test prompts?

We categorized test questions by their stage and relevance to the profile builder:

01
Goal
Investment objectives

02
Timeline
Intended investment duration

03
Budget
Monthly spending capacity

04
Risk
Risk tolerance

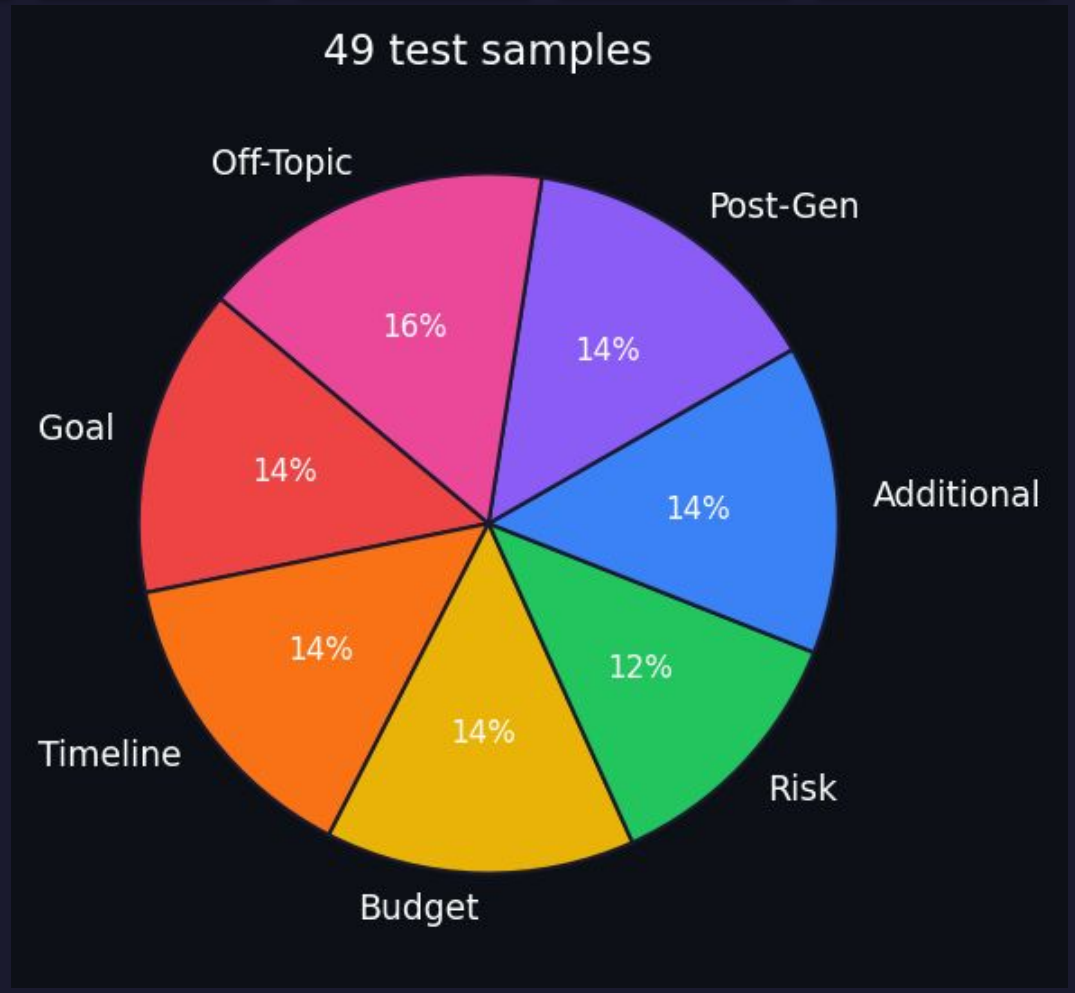
05
Additional Information
Investment-related questions not covered above

06
Post-Gen
Questions asked after portfolio generation

07
Off-Topic
Questions unrelated to any of the above

Test Dataset

49	7	~60%	~20%	~20%
total samples	categories	typical inputs	edge cases	adversarial



Category breakdown

- Goal (7 samples)
- Timeline (7 samples)
- Budget (7 samples)
- Risk (6 samples)
- Additional (7 samples)
- Post-Portfolio Generation (7 samples)
- Off-Topic (8 samples)

Evaluation Metrics

LLM-as-Judge

GPT-4o-mini rates each response against ground truth on a 1–5 scale

- Same rubric applied consistently across all 49 samples
- Secondary metric - used for consistency check and scale
- Catches systematic failures across many inputs quickly

Human Rating

Manual 1-5 rating by team member against ground truth

- Primary metric for pass / fail determination
- Pass threshold: ≥ 4 out of 5
- Catches LLM judge blind spots and context-dependent errors

Rating Rubric

5 - Excellent

Fully matches ground truth

4 - Good

Mostly correct, minor gaps

3 - Adequate

Partially correct

2 - Poor

Mostly fails expectations

1 - Terrible

Irrelevant or contradicts

Prompt Refinements: V1 → V2

1

One-field confirmation loop

Explicit rule: paraphrase understanding and confirm ONE field before saving. Added required format: "It sounds like X - does that sound right?"

2

Edge case handling

Variable budget → ask for steady amount. Numeric risk scale → map to two nearest categories and let user choose. Short timeline → volatility warning. $0 → empathize + suggest $25–50.

3

Post-generation behavioral split

Profile change requests update fields and set profile_changed=true. Informational questions (e.g. "what is VTI?") answer only — never touch profile.

4

Proactive contextual question

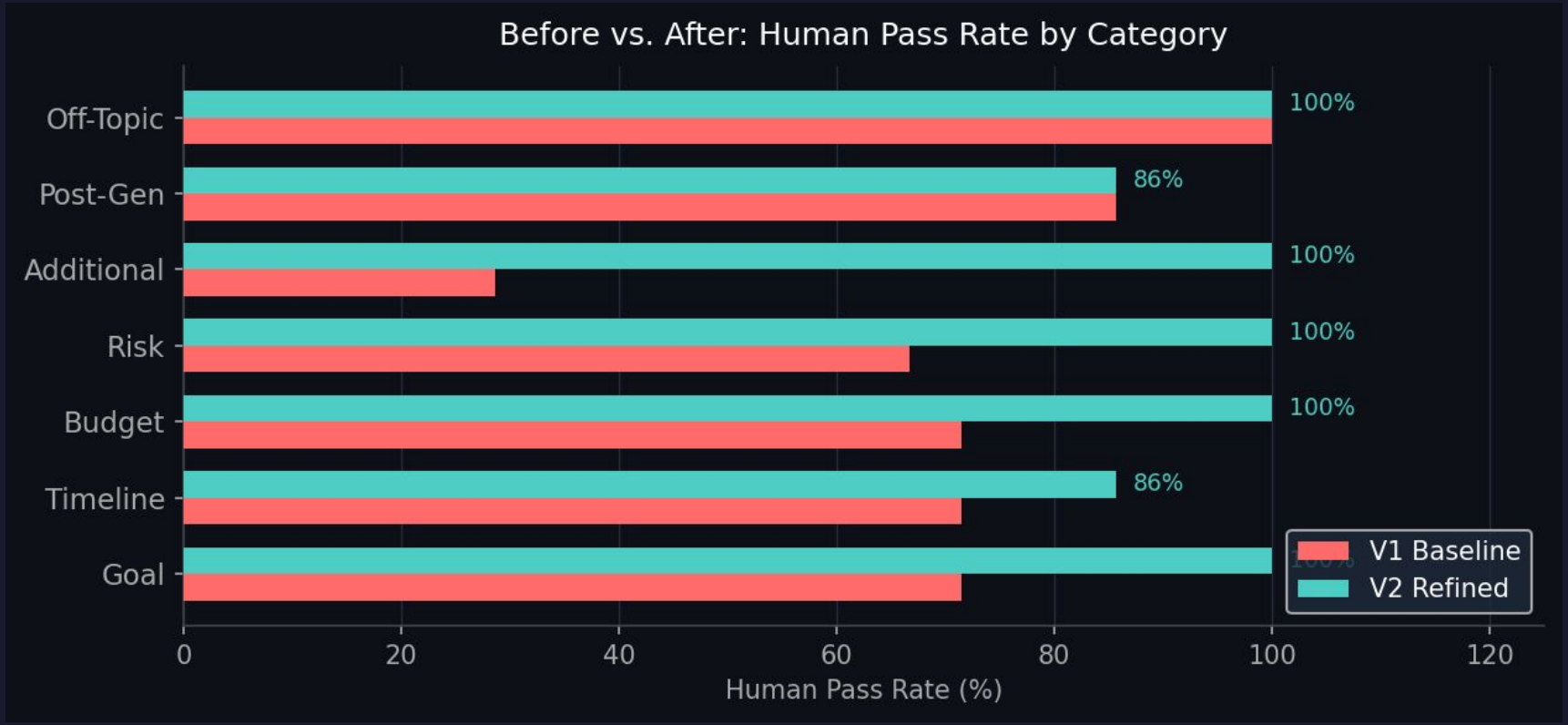
After 4 required fields, ask about existing savings, debt, and life changes instead of generic "anything else?" to surface portfolio-relevant context.

5

Required plain-language risk explanation

Model must describe all three risk levels before asking. Prohibited: numeric confidence outputs like "35% confident."

Before vs. After: Performance Comparison



Metric	V1	V2	Δ
LLM Avg	3.61	4.80	+1.19
Human Avg	3.96	4.76	+0.80
LLM Pass %	67.3%	91.8%	+24.5 pp
Human Pass %	71.4%	95.9%	+24.5 pp

+24.5 pp

human pass rate improvement

71.4% → 95.9%

Conclusion

Qualitative

5 UX friction points identified across a 7-participant think-aloud study. Top issues: financial literacy gap, overly narrow initial prompt framing, and missing regeneration feedback after follow-up messages.

Quantitative

Prompt V2 achieves a 95.9% human pass rate (up from 71.4% baseline, +24.5 pp). LLM-as-judge and human ratings aligned closely (avg delta < 0.1 across all samples).

Combined Approach

Qualitative user observations directly informed the quantitative test cases, enabling evidence-driven prompt iteration from observed friction to measurable reliability improvement.